

Giri's Tech Hub, Pune.

Core Java Machine Test

Batch: June-24 to June-25

Date : 29/05/2026

Time : 08 to 10.00 Am.

Instructions:

1. Solve any 6 questions.
2. Input should be from user.
3. Indentation and comments mandatory.
4. Without using any inbuilt functions.

Q1. Write a java program to check Number Is happy Number or Not happy using function recursion.

Q2. Write a java program to print this pattern.

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5 5 4 3 2 1
```

Q3. Store multiple words and group them according to their length using HashMap<Integer, ArrayList<String>>.

Input Example:

["java","sql","python","c","html"]

Output Example:

1 → [c]

3 → [sql]

4 → [java, html]

6 → [python]

Q4. Store N integers in an ArrayList. Rotate the elements to the right by one position.

Input:

5

10 20 30 40 50

Output:

Rotated List: 50 10 20 30 40

Description:

Store elements in ArrayList.

Move last element to first position manually.

Shift remaining elements one position forward.

Do not use Collections.rotate().

Q5. Store N strings in an ArrayList. Remove duplicate strings manually and store unique strings in another ArrayList.

Input:

java code java list code map

Output:

Unique List: java code list map

Description:

Traverse original ArrayList.

For each element, check manually in second ArrayList using loop comparison.

Do not use contains() method.

Q6. Store N strings in a LinkedHashSet. Remove all palindrome strings.

Input:

level java radar code madam

Output:

Modified Set: java code

Description:

Check each string manually for palindrome (reverse using loop).

Remove palindrome elements using iterator.

Q7. Store N words in a HashMap. Print words that contain at least two vowels.

Input:

java code map idea list

Output:

Words with at least 2 vowels: java code idea

Description:

Store words in HashMap.

Traverse each word manually and count vowels.

Q8. Store N integers in a HashMap. Find the sum of digits of each number and print numbers whose digit sum is even.

Input:

123 45 222 91 84

Output:

Numbers with Even Digit Sum: 123 222 84

-----ALL THE BEST-----